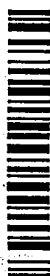




CARIBBEAN EXAMINATIONS COUNCIL
ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE**UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE****PAPER 02***2 hours 30 minutes***05 MAY 2009 (p.m.)****READ THE FOLLOWING INSTRUCTIONS CAREFULLY**

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Candidates **MUST** answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.
5. The use of a silent, non-programmable, scientific calculator is permitted.



MODULE 1

Answer ANY TWO questions.

1. The size of a population of organisms is usually affected by a variety of environmental factors, one of which acts as a limiting factor. Figure 1 shows the population size of an aquatic algal species in an ecosystem that experiences variation in nutrient concentration.

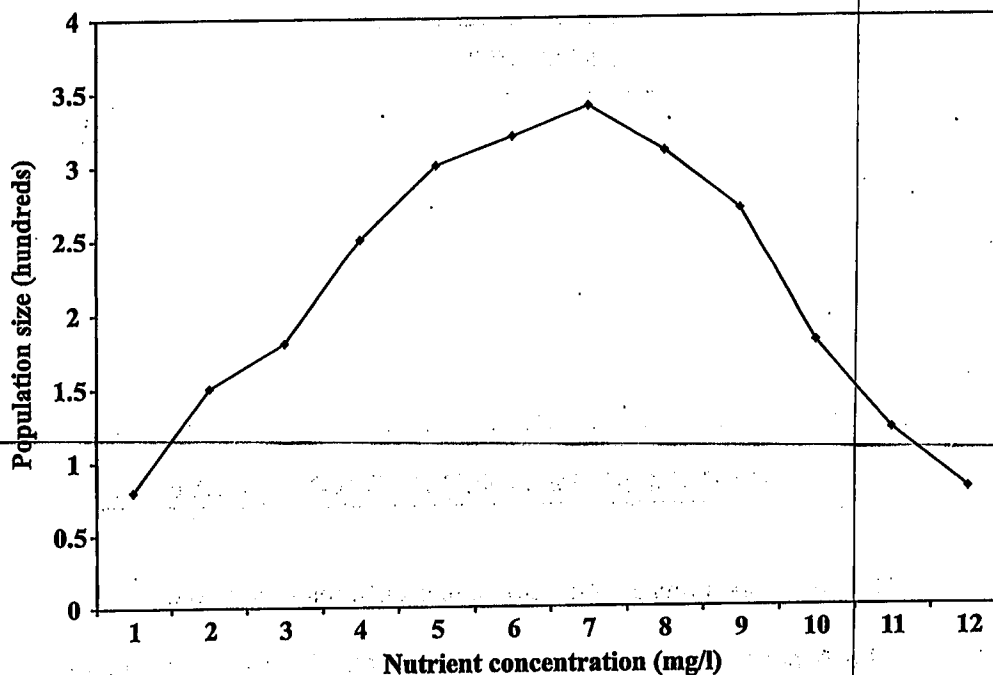


Figure 1. Population size of algal species

- (a) What is meant by the term 'limiting factor'? In your response include THREE ways in which limiting factors operate. [4 marks]
- (b) With reference to Figure 1, describe how the population size of the algal species changes with the variation in nutrients. [4 marks]
- (c) For the relationship between the population size of algae and nutrient concentration represented in Figure 1, explain
- (i) using EIGHT points, how the nutrient variation impacts on the population of algae [8 marks]
 - (ii) using FOUR points, how the variation in the algae population impacts on the population of other species in the ecosystem. [4 marks]

Total 20 marks

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2. (a) Distinguish between 'primary ecological succession' and 'secondary ecological succession'. [3 marks]
- (b) Table 1 shows the current forest area as a percentage of the original forest cover in six different countries.

TABLE 1: FOREST COVER FOR SELECTED COUNTRIES IN 1996

Country	Current Forest Area (% Original Forest Cover)
Belize (Bz)	95.7
Jamaica (Ja)	35.6
Trinidad and Tobago (TT)	35.2
Martinique (Mq)	92.4
Haiti (Ht)	0.8
St Lucia (SL)	51.5

- (i) An insert is provided for this question.

On the axes provided, illustrate the data shown in Table 1 as a bar chart.

[5 marks]

- (ii) Use the information presented in Table 1 to answer the following questions:

- a) Identify the country where secondary ecological succession is

– MOST likely to occur B [1 mark]

– LEAST likely to occur. H [1 mark]

- b) Provide FIVE reasons to support your answer in EACH case in (ii) a) above. [10 marks]

Total 20 marks

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3. Snail kites are birds that feed on freshwater snails. Figure 2 shows the variation in population numbers of both snails and snail kites in an ecosystem for the first eight-month period, January to August, of the year.

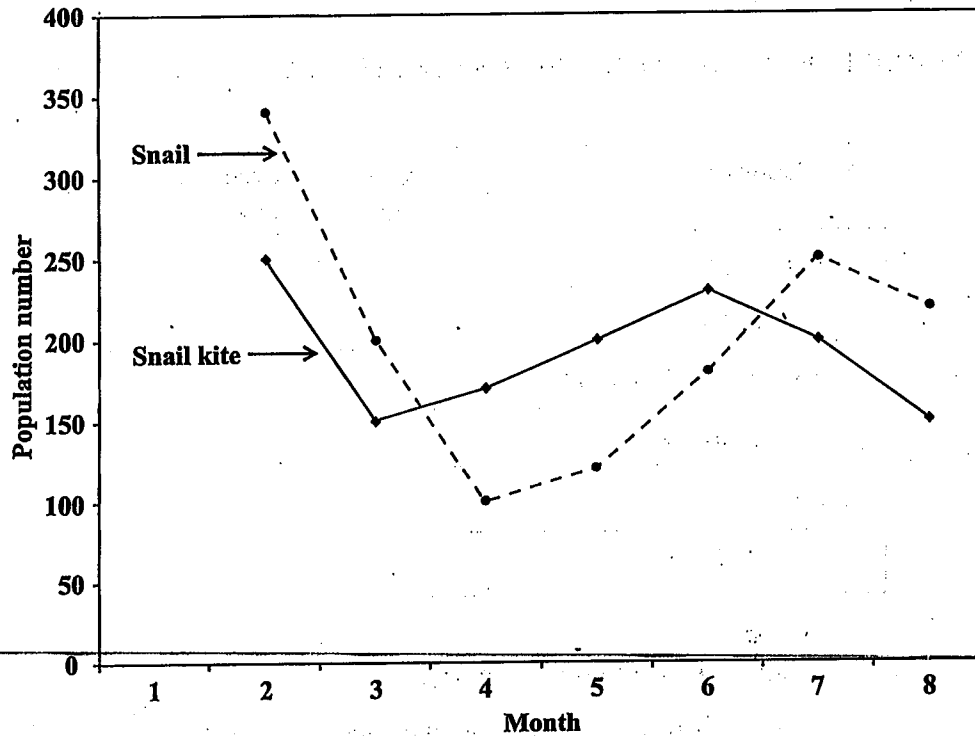


Figure 2. Variation in population size

- (a) With reference to Figure 2, describe the variation in the population size of BOTH organisms over the eight-month period. [4 marks]
- (b) Outline how predation may be of benefit to the prey population represented in Figure 2. [4 marks]
- (c) Explain how ONE density independent factor may impact the size of the snail population. [4 marks]
- (d) Assess TWO possible impacts on the ecosystem if the population of snails is significantly reduced by draining the waterway. [8 marks]

Total 20 marks

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MODULE 2

Answer ANY TWO questions.

4. (a) Describe the relationship between per capita GDP and the level of poverty existing in countries. [4 marks]
- (b) The graph in Figure 3 shows the relationship between total fertility rate and per capita gross domestic product (GDP) for some countries.

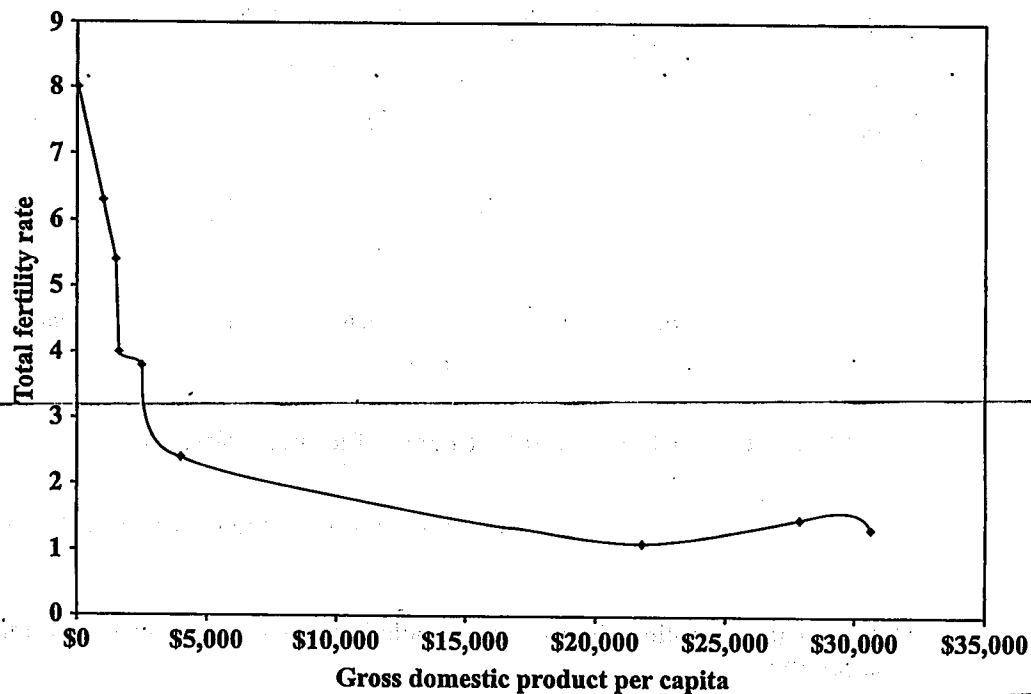


Figure 3. Total fertility rate versus per capita GDP

- (b) (i) Outline the relationship between total fertility rate and per capita GDP based on the graph in Figure 3. [4 marks]
- (ii) Explain how the level of poverty in countries accounts for the relationship outlined in (b) (i) above. [12 marks]

Total 20 marks

5. Figure 4 shows the death rate (DR) and birth rate (BR) of a developing country from the year 1900 to 2000.

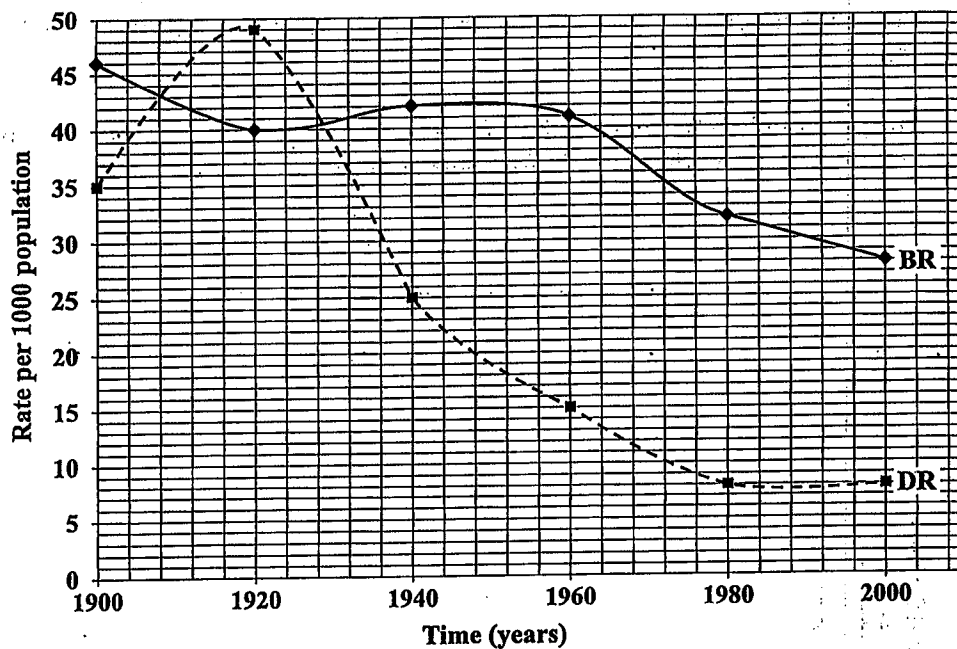


Figure 4. Death rate and birth rate of a developing country

- (a) Describe how the growth rate of the population changed between 1908 and 2000. [4 marks]
- (b) Calculate the doubling time of the population if its rate of growth remains constant beyond 2000. [6 marks]
- (c) Outline TWO factors that may be responsible for the changes in the growth rate of the population since 1928. [4 marks]
- (d) Assess the role of government policies in controlling population growth rate. Include SIX points in your response. [6 marks]

Total 20 marks

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6. Table 2 presents data for water consumption per capita for developing and developed countries.

TABLE 2: WATER CONSUMPTION PER CAPITA

Group of Countries	Water Consumption per Capita (m ³)
Developing	545
Developed	948

- (a) Calculate the percentage difference in the per capita water consumption between developing and developed countries. [4 marks]
- (b) Describe an environmental impact associated with high per capita water consumption. [4 marks]
- (c) Account for the difference in per capita water consumption between developing and developed countries. [4 marks]
- (d) A student wrote the following statement based on the information presented in Table 2.

"The high consumption patterns of developed countries are responsible for the world's environmental degradation."

Assess the statement in light of current world population distribution. [8 marks]

Total 20 marks

MODULE 3

Answer ANY TWO questions.

7. The Government of a developing Caribbean country has announced that after many years of exploration, major deposits of gold and gravel were discovered in a largely remote and forested area of the country. Figure 5 presents the results of research conducted by the Government on the environmental impacts of mining these two natural resources.

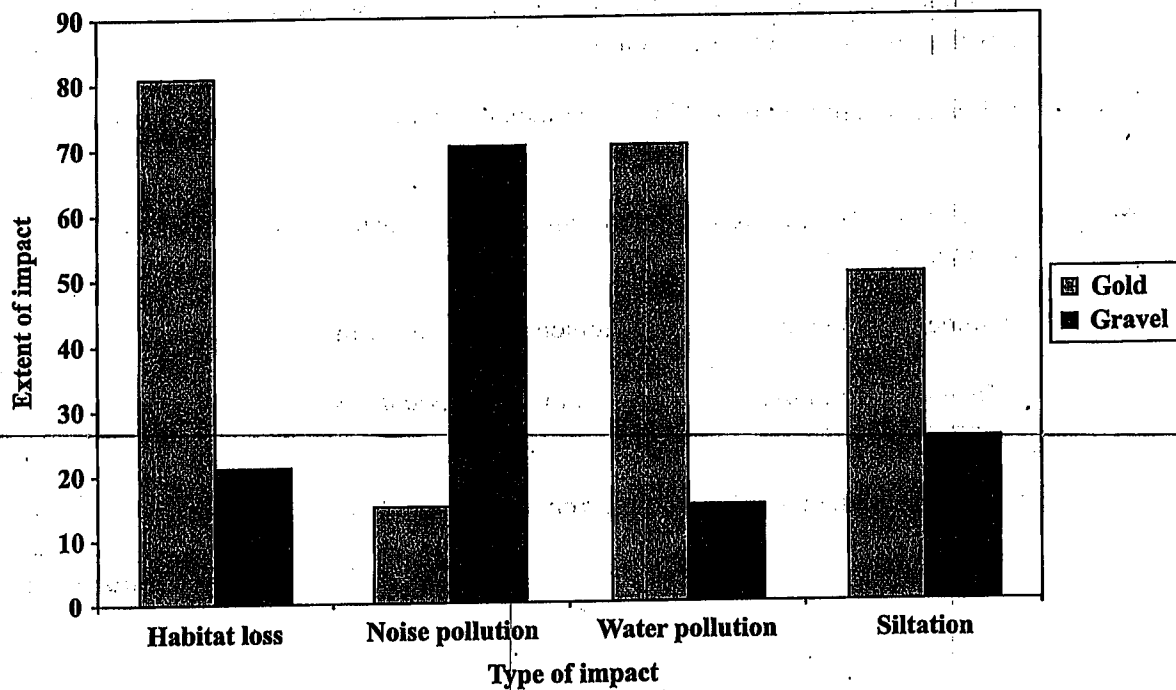


Figure 5. Environmental impact of mining

- (a) Based on the results depicted in Figure 5, compare the environmental impact of extracting these two resources. Include FOUR points in your response. [4 marks]
- (b) Account for the difference in the level of noise pollution when these two resources are extracted. Include FOUR points in your response. [4 marks]
- (c) Discuss THREE factors (excluding environmental) that the Government should consider before permitting the extraction of these two resources. [12 marks]

Total 20 marks

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8. Table 3 presents information regarding the impact on mangrove ecosystems of a restricted freshwater flow and the discharge of sewage at three locations, A, B and C.

TABLE 3: IMPACT ON MANGROVE ECOSYSTEM

Location	Death of Mangroves (Percentage)	Freshwater Flow through Mangrove Ecosystem	Sewage Discharge in Mangrove Ecosystem
A	0	Yes	Yes
B	0	Yes	No
C	70	No	Yes

✓ Control.

- (a) Describe ONE function of mangrove ecosystems. [4 marks]
- (b) (i) What can you deduce from the information presented in Table 3? [2 marks]
- (ii) Justify your response in (b) (i) above. [2 marks]
-
- (c) Describe the effect that EACH of the following could have on mangrove ecosystems. Include SIX points in EACH response. *3 lines.*
- (i) Reduced freshwater flow through mangrove ecosystems [6 marks]
- (ii) Discharge of sewage in mangrove ecosystems [6 marks]

Total 20 marks

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9. Figure 6 shows the percentage of countries marketing both specific ecotourism destinations and traditional tourism destinations between 1985 and 2005.

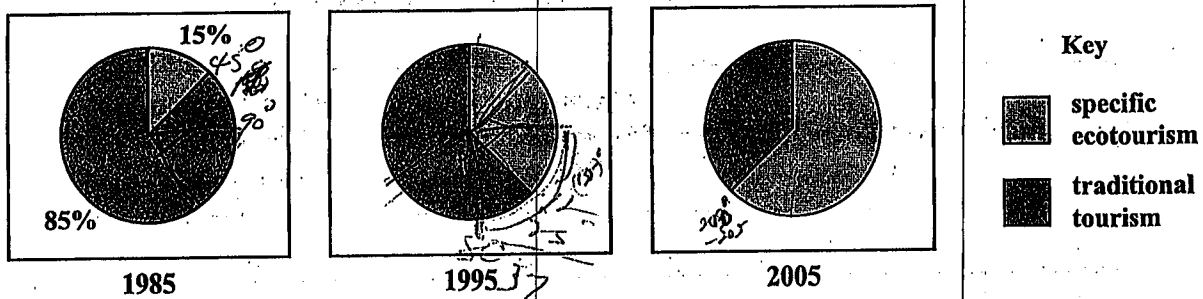


Figure 6. Changing trend in tourism travel market

- (a) With reference to Figure 6, comment on the trend in the tourism travel market from 1985 to 2005. [4 marks]
① Demand Change ② Advertisement ③ Education ④ Cheaper
- (b) State FOUR possible reasons for the trend shown in Figure 6. [4 marks]
- (c) Discuss THREE ways in which ecotourism can be used as a tool for the conservation of natural resources. [12 marks]

→ high demand for ecotourism, will be increase demand for natural resource is you can not deplete it.

→ contains income without depleting or removing resources.

Total 20 marks

END OF TEST

→ Education of natural resource management & biodiversity.

Olympus
 styles \$50 SW
 shock + water proof

15-37

25

27
 15
 12

Barbados
 Greene hall
 Ocean Park
 native Sanctuary.